

WEEK 2 TASK

VIRTUAL SECURITY INCIDENT RESPONSE SIMULATION

Hands-on Internship Report | Controlled Virtual Environment

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Domain: Cybersecurity / SOC / Incident Response

Framework: NIST-aligned Incident Response Lifecycle

Environment: Windows Virtual Machine

Exercise type: Controlled security simulation

1. Executive Summary

This report documents a hands-on incident-response simulation performed in a controlled Windows virtual machine. A harmless simulated endpoint incident was created to practice detection, analysis, evidence preservation, containment, eradication, recovery and validation. No real malware, production system or unauthorized external target was involved.

2. Objective

The objective was to identify the simulated breach, determine its scope, collect evidence, document containment, remove simulated persistence, recover the endpoint and recommend controls that reduce the chance and impact of similar incidents.

3. Lab Setup

- Windows virtual machine used as the affected endpoint.
- Administrator PowerShell used for investigation and evidence collection.
- C:\IR-Lab used as the local evidence directory.
- Process, network, scheduled-task, SHA-256 and Windows event evidence collected.
- All activity was intentionally harmless and confined to the virtual lab.

4. Simulated Incident Scenario

A phishing-style endpoint compromise was simulated without malware. The harmless file dropper_sim.txt represented a suspicious payload and the one-time UpdateCheck-Simulation scheduled task represented persistence. The analyst then investigated the indicators, preserved evidence, documented containment, removed the simulated persistence and artifact, and validated the endpoint.

5. Attack Chain

Simulated phishing → user execution → simulated artifact → scheduled-task persistence → detection → investigation → evidence preservation → containment → eradication → recovery → validation.

6. Indicators of Compromise

Indicator	Evidence	Meaning
dropper_sim.txt	C:\IR-Lab\dropper_sim.txt	Harmless simulated payload
UpdateCheck-Simulation	Windows scheduled task	Simulated persistence
SHA-256 hash	artifact_hash.txt	Artifact identification/integrity
Process information	processes.txt	Endpoint activity evidence
Network connections	network_connections.txt	Network investigation evidence
Security/System events	security_events.txt and system_events.txt	Timeline correlation

7. Hands-on Activities Completed

1. Created and inspected the harmless simulated artifact.
2. Created and verified simulated scheduled-task persistence.
3. Collected process and PowerShell process information.
4. Collected active network connection information.
5. Calculated a SHA-256 hash.
6. Exported recent Windows Security and System events.
7. Created a response timeline and evidence manifest.
8. Documented containment, eradication and recovery.
9. Performed final validation.

8. Evidence Screenshots

```
PS C:\Windows\system32> New-Item -ItemType Directory -Force C:\IR-Lab | Out-Null
PS C:\Windows\system32> Set-Content C:\IR-Lab\dropper_sim.txt "SIMULATED INCIDENT ARTIFACT - NO MALWARE"
PS C:\Windows\system32> Get-Item C:\IR-Lab\dropper_sim.txt

Directory: C:\IR-Lab

Mode                LastWriteTime         Length Name
----                -
-a---             16-09-2026      20:06           42 dropper_sim.txt

PS C:\Windows\system32> $action = New-ScheduledTaskAction -Execute "cmd.exe" -Argument "/c echo SIMULATED-PERSISTENCE > C:\IR-Lab\persistence_check.txt"
>> $trigger = New-ScheduledTaskTrigger -Once -At (Get-Date).AddMinutes(2)
>> Register-ScheduledTask -TaskName "UpdateCheck-Simulation" -Action $action -Trigger $trigger -Force

TaskPath              TaskName              State
-----
\                      UpdateCheck-Simulation Ready

PS C:\Windows\system32> Get-ScheduledTask -TaskName "UpdateCheck-Simulation"

TaskPath              TaskName              State
-----
\                      UpdateCheck-Simulation Ready
```

Figure 1 — Simulated artifact and scheduled-task creation.

The screenshot documents creation of a harmless simulated artifact and the UpdateCheck-Simulation scheduled task inside the Windows VM. This represents the simulated payload and persistence stage.

```
Windows PowerShell
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Try the new cross-platform PowerShell https://aka.ms/pscore6

PS C:\Windows\system32> New-Item -ItemType Directory -Force C:\IR-Evidence
>> New-Item -ItemType Directory -Force C:\IR-Evidence\Screenshots
>> New-Item -ItemType Directory -Force C:\IR-Evidence\Logs
>> New-Item -ItemType Directory -Force C:\IR-Evidence\Artifacts

Directory: C:\

Mode                LastWriteTime         Length Name
----                -
d-----         16-09-2026      20:01             IR-Evidence

Directory: C:\IR-Evidence

Mode                LastWriteTime         Length Name
----                -
d-----         16-09-2026      20:01             Screenshots
d-----         16-09-2026      20:01             Logs
d-----         16-09-2026      20:01             Artifacts

PS C:\Windows\system32>
```

Figure 2 — Preserved investigation evidence.

The C:\IR-Lab directory contains process, network, scheduled-task and hash evidence plus containment and recovery records. These files support the investigation timeline.

```
PS C:\Windows\system32> Test-Path C:\IR-Lab\dropper_sim.txt
False
PS C:\Windows\system32> Get-ScheduledTask -TaskName "UpdateCheck-Simulation" -ErrorAction SilentlyContinue
PS C:\Windows\system32> Get-Process powershell,pwsh -ErrorAction SilentlyContinue | Select-Object Name,Id

Name            Id
----            --
powershell      13036

PS C:\Windows\system32> Get-WinEvent -LogName Security -MaxEvents 30 |
>> Select-Object TimeCreated,Id,ProviderName,Message |
>> Out-File C:\IR-Lab\security_events.txt
PS C:\Windows\system32> Get-WinEvent -LogName System -MaxEvents 30 |
>> Select-Object TimeCreated,Id,ProviderName,Message |
>> Out-File C:\IR-Lab\system_events.txt
PS C:\Windows\system32> Get-ChildItem C:\IR-Lab

Directory: C:\IR-Lab

Mode                LastWriteTime         Length Name
----                -
-a---          16-09-2026   20:20           1154 artifact_hash.txt
-a---          16-09-2026   20:15              0 CONTAINMENT_COMPLETED.txt
-a---          16-09-2026   20:16            94 CONTAINMENT_NOTE.txt
-a---          16-09-2026   20:19          17474 network_connections.txt
-a---          16-09-2026   20:17          75854 processes.txt
-a---          16-09-2026   21:35           116 RECOVERY_COMPLETED.txt
-a---          16-09-2026   20:19          1570 scheduled_task.txt
-a---          16-09-2026   21:38          12174 security_events.txt
-a---          16-09-2026   21:38          12174 system_events.txt
```

Figure 3 — Eradication and final validation.

The artifact check returned False and the scheduled-task query returned no result, confirming removal of the simulated artifact and persistence. The remaining PowerShell process is the analyst's own administrative session.

```

PS C:\Windows\system32> @"
>> WEEK 2 INCIDENT RESPONSE - EVIDENCE MANIFEST
>>
>> Environment: Controlled Windows Virtual Machine
>> Incident: Simulated endpoint compromise
>> Evidence collected:
>> - Process information
>> - Network connections
>> - Scheduled task information
>> - SHA-256 artifact hash
>> - Windows Security events
>> - Windows System events
>> - Incident timeline
>> - Containment documentation
>> - Recovery documentation
>>
>> Final status:
>> Simulated artifact removed.
>> Simulated persistence removed.
>> Validation completed.
>> No real malware or production system was involved.
>> "@ | Set-Content C:\IR-Lab\EVIDENCE_MANIFEST.txt
PS C:\Windows\system32> Get-Content C:\IR-Lab\EVIDENCE_MANIFEST.txt
WEEK 2 INCIDENT RESPONSE - EVIDENCE MANIFEST

Environment: Controlled Windows Virtual Machine
Incident: Simulated endpoint compromise
Evidence collected:
- Process information
- Network connections
- Scheduled task information
- SHA-256 artifact hash
- Windows Security events
- Windows System events
- Incident timeline
- Containment documentation
- Recovery documentation

Final status:
Simulated artifact removed.
Simulated persistence removed.
Validation completed.

```

Figure 4 — Evidence manifest.

The manifest records the controlled Windows VM, simulated endpoint compromise, collected evidence and final status. It explicitly confirms that no real malware or production system was involved.

```

PS C:\Windows\system32> @"
>> INCIDENT RESPONSE TIMELINE
>>
>> 14:00 - Lab preparation completed.
>> 14:05 - Suspicious simulated activity identified.
>> 14:07 - Simulated artifact and persistence mechanism investigated.
>> 14:10 - Affected virtual workstation placed in simulated containment.
>> 14:15 - Process, network, task and artifact evidence preserved.
>> 14:25 - Simulated persistence and artifact removed.
>> 14:35 - Recovery procedure documented.
>> 14:45 - Final validation completed.
>> 15:00 - Post-incident analysis and recommendations prepared.
>> "@ | Set-Content C:\IR-Lab\incident_timeline.txt
PS C:\Windows\system32> Get-Content C:\IR-Lab\incident_timeline.txt
INCIDENT RESPONSE TIMELINE

14:00 - Lab preparation completed.
14:05 - Suspicious simulated activity identified.
14:07 - Simulated artifact and persistence mechanism investigated.
14:10 - Affected virtual workstation placed in simulated containment.
14:15 - Process, network, task and artifact evidence preserved.
14:25 - Simulated persistence and artifact removed.
14:35 - Recovery procedure documented.
14:45 - Final validation completed.
15:00 - Post-incident analysis and recommendations prepared.
PS C:\Windows\system32> Write-Host "=== ARTIFACT CHECK ==="
>> Test-Path C:\IR-Lab\dropper_sim.txt
>>
>> Write-Host "=== PERSISTENCE CHECK ==="
>> Get-ScheduledTask -TaskName "UpdateCheck-Simulation" -ErrorAction SilentlyContinue
>>
>> Write-Host "=== EVIDENCE FILES ==="
>> Get-ChildItem C:\IR-Lab | Select-Object Name,Length
=== ARTIFACT CHECK ===
False
=== PERSISTENCE CHECK ===
=== EVIDENCE FILES ===

Name                               Length
----
artifact_hash.txt                  1154
CONTAINMENT_COMPLETED.txt          0

```

Figure 5 — Incident timeline and validation.

The timeline documents preparation, detection, investigation, containment, evidence preservation, eradication, recovery and final validation.

9. Incident Response Process

Preparation

Create an isolated VM, establish an evidence directory, verify logging and maintain a known-good recovery point.

Detection & Analysis

Review processes, PowerShell activity, files, scheduled tasks, network connections and Windows events; correlate evidence into a timeline.

Containment

Document isolation of the affected virtual workstation and protect evidence before cleanup. In production, approved EDR, firewall or NAC controls would be used.

Eradication

Unregister the simulated scheduled task and remove the harmless simulated artifact. Verify that both are absent.

Recovery

Document restoration to a known-good state, apply baseline controls and validate the endpoint before normal use.

Post-Incident Activity

Record root cause, control gaps, lessons learned and corrective actions; update the response playbook.

10. Evidence Preservation

- processes.txt — process evidence.
- network_connections.txt — network evidence.
- scheduled_task.txt — persistence evidence.
- artifact_hash.txt — SHA-256 evidence.
- security_events.txt — Windows Security events.
- system_events.txt — Windows System events.
- incident_timeline.txt — chronological response record.
- EVIDENCE_MANIFEST.txt — evidence inventory and final status.

11. Post-Incident Analysis

Security gap	Potential impact	Recommended improvement
Limited centralized logging	Slower cross-source investigation	Centralize endpoint/authentication/network logs in a SIEM.
Limited application control	Unapproved scripts/artifacts may execute	Use application control and least privilege.
Network reachability	Could increase incident scope	Segment networks and restrict unnecessary egress.
Credential protection	Could enable unauthorized access	Use unique credentials, MFA where applicable and rotation.
Untested response procedures	Inconsistent response under pressure	Maintain playbooks and conduct recurring exercises.

12. Lessons Learned

- Collect evidence before destructive remediation when investigation requires it.
- Correlating process, file, task, network and event data produces a clearer timeline.
- Controlled simulations allow incident-response skills to be practiced safely.
- Known-good VM snapshots support rapid recovery.
- Findings should become tracked corrective actions.

13. Final Validation

The final artifact check returned False, confirming that dropper_sim.txt had been removed. The scheduled-task query returned no result, confirming removal of UpdateCheck-Simulation. The PowerShell process visible in the validation evidence is the analyst's active administrative session and is not treated as malicious persistence.

14. Corrective Action Plan

- Centralize security telemetry in a SIEM.
- Enable appropriate PowerShell and endpoint logging.
- Deploy endpoint detection and response controls.
- Apply least privilege and application control.
- Segment networks and restrict unnecessary outbound traffic.
- Use MFA and unique privileged credentials where applicable.
- Maintain golden VM snapshots and rebuild procedures.
- Run recurring tabletop and technical incident-response exercises.

15. Conclusion

The exercise demonstrated the complete incident-response lifecycle in a safe virtual environment. Evidence was collected and preserved, the simulated incident was contained, the simulated persistence and artifact were eradicated, and final validation confirmed their removal. The post-incident review identified practical improvements in logging, endpoint protection, access control, network segmentation and response readiness.

16. Scope and Integrity Note

This document describes a training simulation. The artifacts were intentionally harmless and created only to demonstrate incident-response procedures. It must not be represented as a real production breach.

17. References

NIST SP 800-61 Rev. 3 — Incident Response Recommendations and Considerations for Cybersecurity Risk Management (2025).
NIST Incident Response Project.
CISA Cybersecurity Incident Response Playbooks.